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LAB ASSIGNMENT - 02

★ PROBLEM STATEMENT

Write a prog. in Java having hierarchical inheritance with base class as Employee & derived classes as Full Time Employee & InternEmployee with methods DisplaySalary is base class & Calculate salary is derived. Calculate will calculate as per increment given to full time & intern Employee.

Full time → 50% hike

Intern → 25% hike

★ OBJECTIVE

1) To study inheritance in Java
2) To study why use inheritance
3) To study types of inheritance.

★ THEORY

(1) What is inheritance in Java?

(1) Inheritance in Java is an object oriented concept where one class acquires the prop and behaviours of another class using the extends keyword. It promotes code reuse & hierarchy.

(2) Why use inheritance?

(2) Used to reuse existing code, reduce redundancy and improve maintainability. It also supports method overriding, enabling runtime polymorphism & better program flexibility.

(3) Types of Inheritance.

(3) Java supports single, multi-level & Hierarchical inheritance.
Multiple is not supported using classes but can be achieved through interfaces.

* CONCLUSION : Thus, we have successfully implemented usage of inheritance.

* FAQ

(1) Is multiple inheritance supported in Java?

(1) Does not support Multiple using classes to avoid ambiguity. However it is achieved using interfaces, where a class can implement multiple interfaces.

(2) What is Is-A relationship?

(2) Represents inheritance, where a subclass is a type of its superclass. It is implemented using the

extends keyword & supports code reuse.

(3) Are constructors & instance initialization blocks inherited?

(3) No, not inherited by subclasses. However, a subclass can call the superclass constructor using the super() keyword.